

Three ways an LLM looked better than it was

Removing one confound at a time from an LLM-guided quantum circuit search

Headline: each apparent advantage came from something other than design skill - circuit size, then a bad starting point. The LLM beats a textbook ansatz on all four tasks and never beats random search on any (30 seeds).

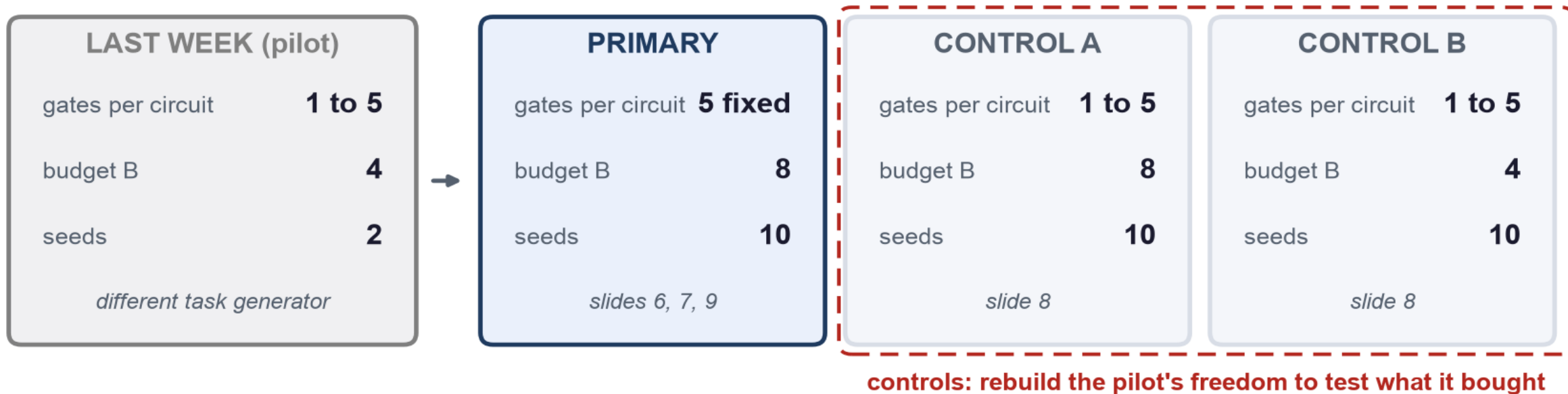
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WHAT CHANGED

Last week the LLM could choose how big the circuit was



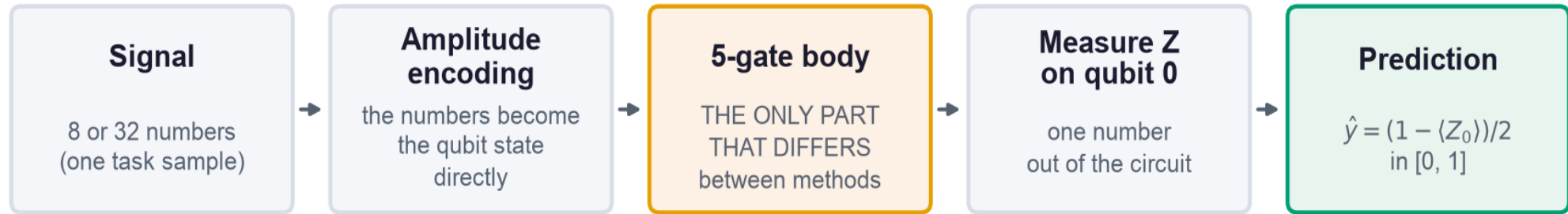
Longer circuits usually score better. So a proposer that always asks for the maximum length wins without designing anything. Fixing the length takes that move away.

Every rule, stated so you can check any circuit by eye

Constraint	Value
Tasks	T1 peak position · T2 sinusoid frequency · T3 change-point location · T4 one peak vs two (binary)
Qubits	3 for T1, T2 · 5 for T3, T4 (a circuit on n qubits holds 2^n numbers)
Gate count	5 exactly (PRIMARY) · 1 to 5 (both controls)
Gate set	H, RX, RY, RZ (one qubit) · CRX, CRY, CRZ (two qubits)
Angles	one number per gate, anywhere in [-pi, pi] (H has none)
Budget B	8 distinct circuits scored per method (4 in Control B) · 10 seeds
Training	none - proposed angles are used exactly as written
Encoding / readout	amplitude encoding; measure Z on qubit 0
Classical params	zero - nothing outside the circuit can fix a bad circuit

One operation = one physical gate; a 5-gate circuit compiles to 5 gates. The same grammar check is applied to classical samplers and LLM replies alike

One number: RMSE, the typical size of the error



No training anywhere: the angles a method proposes are used exactly as written. No classical layers: nothing outside the circuit can fix a bad circuit.

Validation RMSE

drives the search - each method keeps its own best

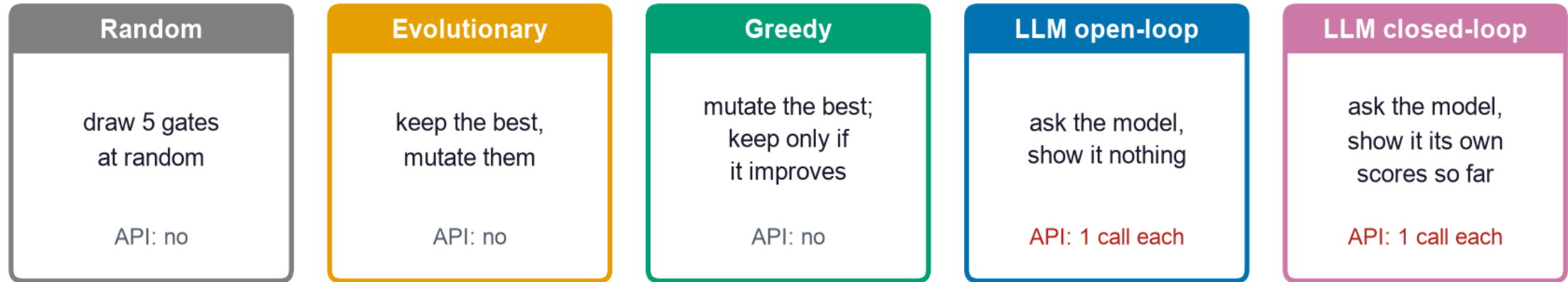
Test RMSE

reported once, after the choice is frozen. Never used to choose.

T4 is yes/no

so RMSE there is the error on a 0/1 label. AUROC is shown too - slide 7.

Five methods, one shared budget



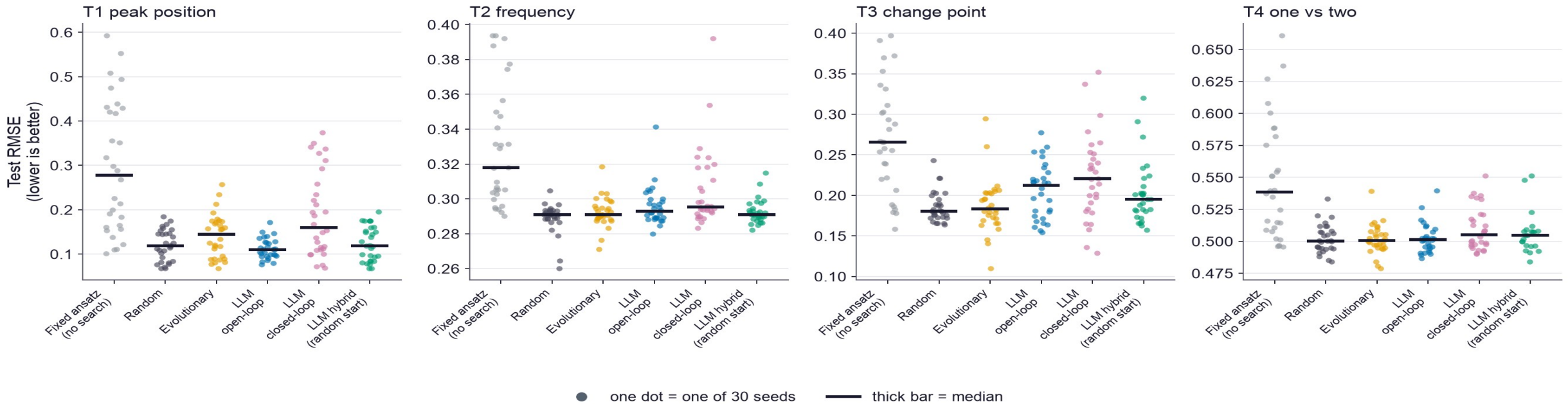
All five get the same budget B: the same number of DISTINCT circuits actually scored. Repeats and malformed proposals are free but earn nothing.

Open-loop vs closed-loop: open-loop never sees how its own circuits scored; closed-loop does, eight times. That is the only difference between them.

RESULT

16 gates, 5 qubits, $B = 8$, 30 seeds all four tasks

The LLM beats the textbook ansatz, never beats random

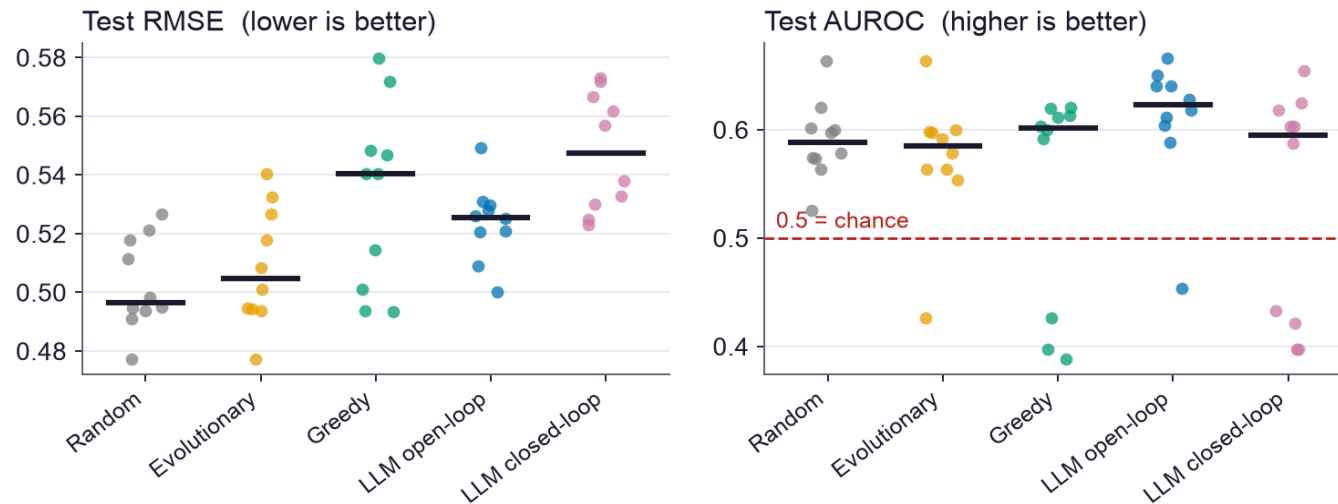


Every panel has its own y-axis. RMSE compares arms inside a panel, never across panels.

vs the fixed ansatz: significant on all 4 tasks. **vs random: no difference detected on any.** Random holds the better median on T2, T3 and T4 (T3 0.180 vs 0.212); the LLM leads only on T1 (0.110 vs 0.119), and not significantly.

On T4 the metric, not the method, picks the winner

T4 is binary classification: RMSE against a 0/1 label is $\sqrt{\text{Brier}}$, a proper score, but it mixes calibration with discrimination.



AUROC

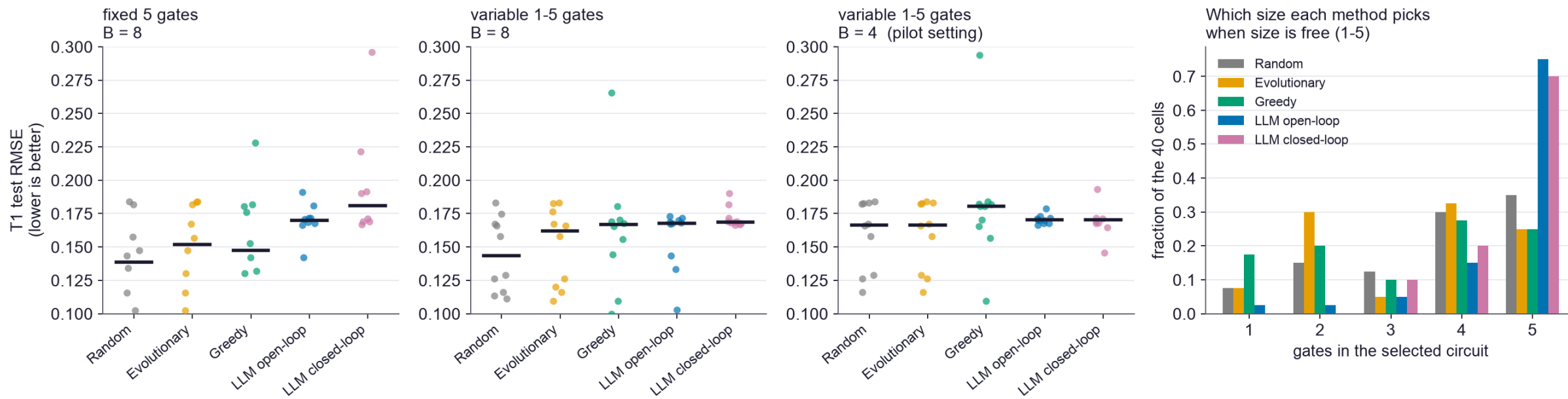
= how often the model ranks a two-peak signal above a one-peak one. 1.0 perfect, 0.5 coin flip.

Why they disagree

RMSE also punishes outputting 0.45 / 0.55 instead of 0 / 1. AUROC only cares about the order.

Neither reading is strong: AUROC spans 0.59 to 0.62 against 0.5 for chance. Five gates without training barely solve T4 at all.

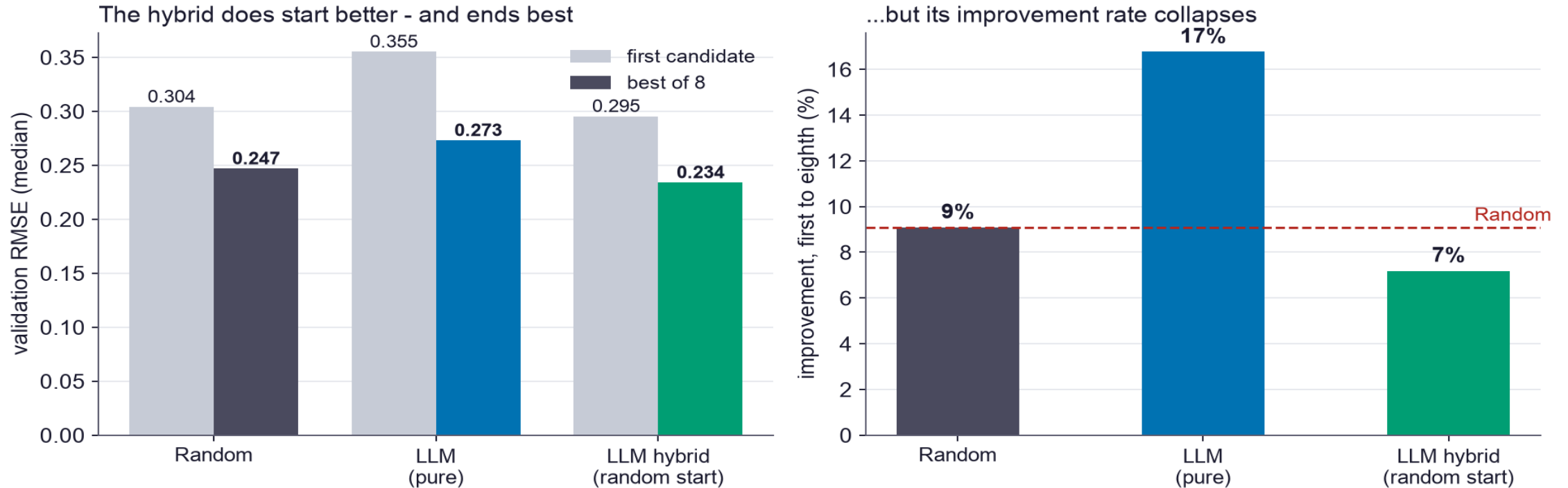
Last week's advantage was a size decision, not a design one



Left three panels: one dot = one of 10 seeds, thick bar = median. Only Random degrades as circuit size becomes a free choice, because the LLM almost always picks the maximum size while Random draws it uniformly. The pilot rewarded that single decision.

Free the length and only Random gets worse (0.139 → 0.167); the LLM does not move (0.170 → 0.170), because it was already asking for 5. At the pilot's setting they meet.

The LLM's "refinement skill" was the room a bad start leaves



The pure LLM improved 17% where random improved 9%, which looked like refinement skill. Give it a good start and that rate falls to 7%, below random. The 17% was the room a bad start leaves, not an ability.

I built the hybrid arm expecting it to win. It reached the best final score of any arm and still could not beat random - because the ability it was built to exploit was not there.

What I can and cannot claim

Supported

- The LLM beats a textbook ansatz on all 4 tasks
- It never beats random search - 30 seeds, 12 pre-declared contrasts
- Given free length it picks the maximum ~70% of the time
- Its apparent refinement skill vanishes given a good start

NOT supported

- "LLMs cannot design circuits" - one small model, $B = 8$
- "Task context does not help" - never A/B tested at equal seeds
- Anything about hardware - this is exact simulation
- Comparing RMSE across different tasks

Next: give the LLM a decision that is actually hard, and check any advantage the same way.